



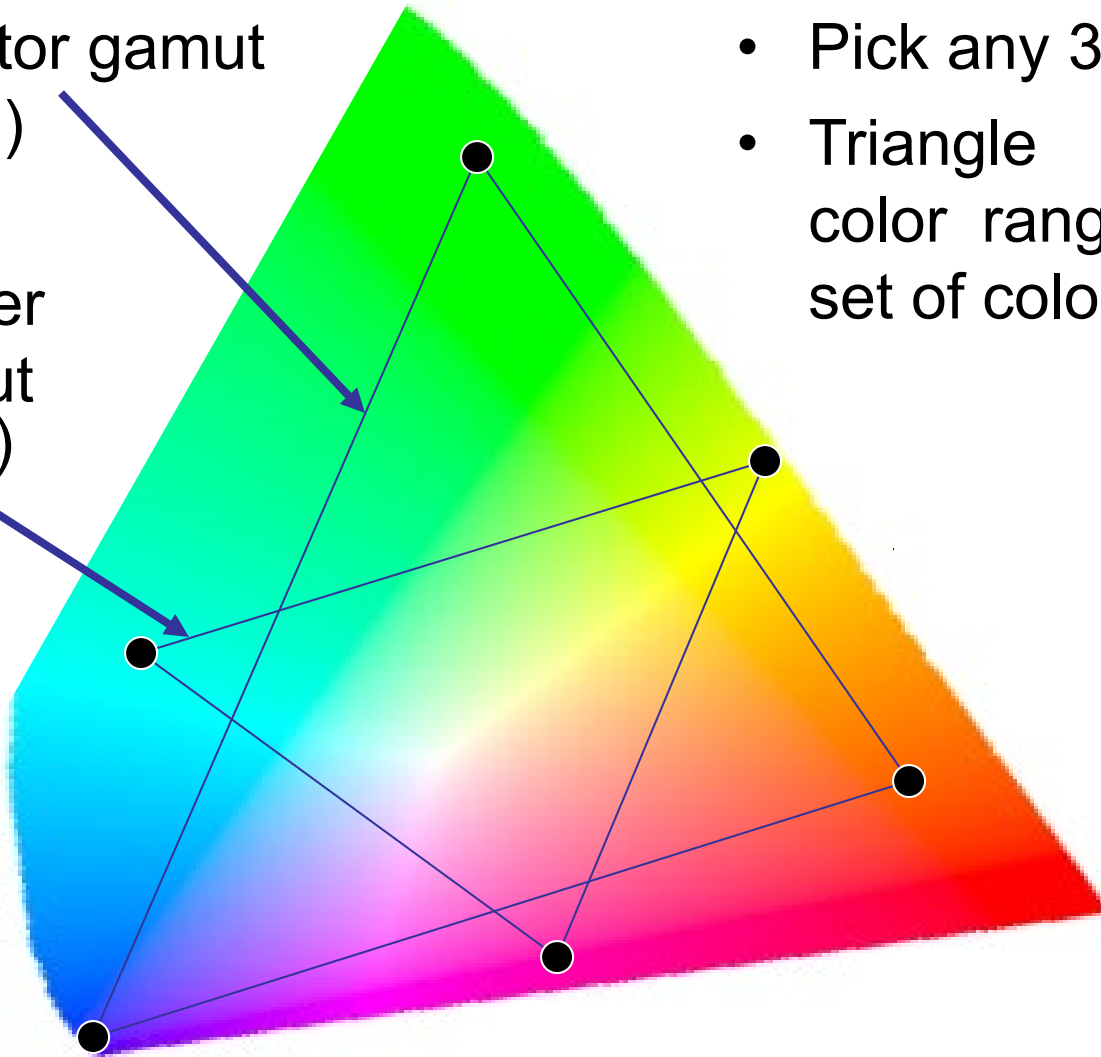
Set Operations

ค่ายโอลิมปิกวิชาการ วิชาคอมพิวเตอร์ ค่ายที่ 1 วันที่ 5
วิทยากร - ผู้ช่วยศาสตราจารย์ ดร. อัครา ประโยชน์

Sets of Colors

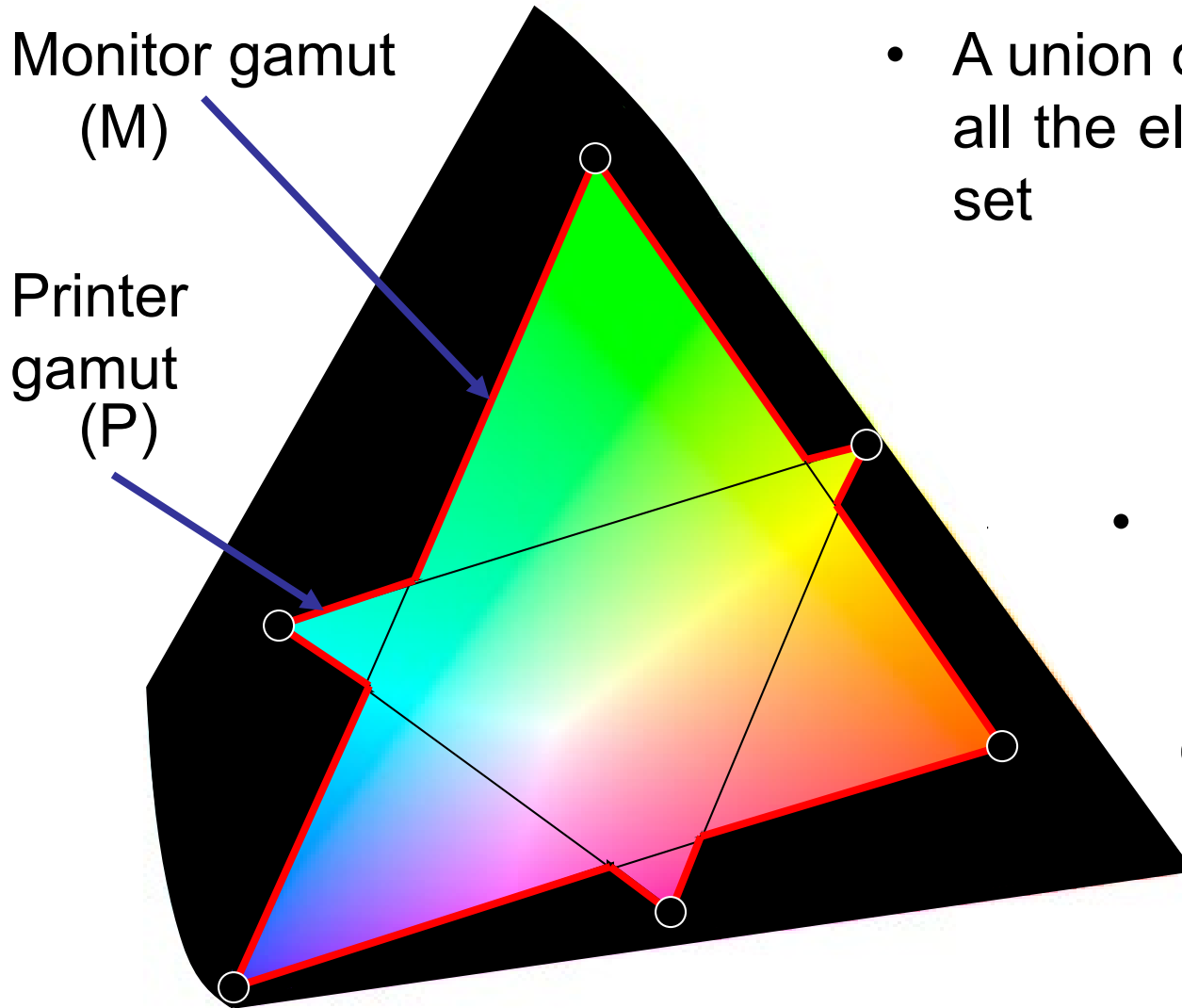
Monitor gamut
(M)

Printer
gamut
(P)



- Pick any 3 “primary” colors
- Triangle shows mixable color range (gamut) – the set of colors

Set operations: Union 1

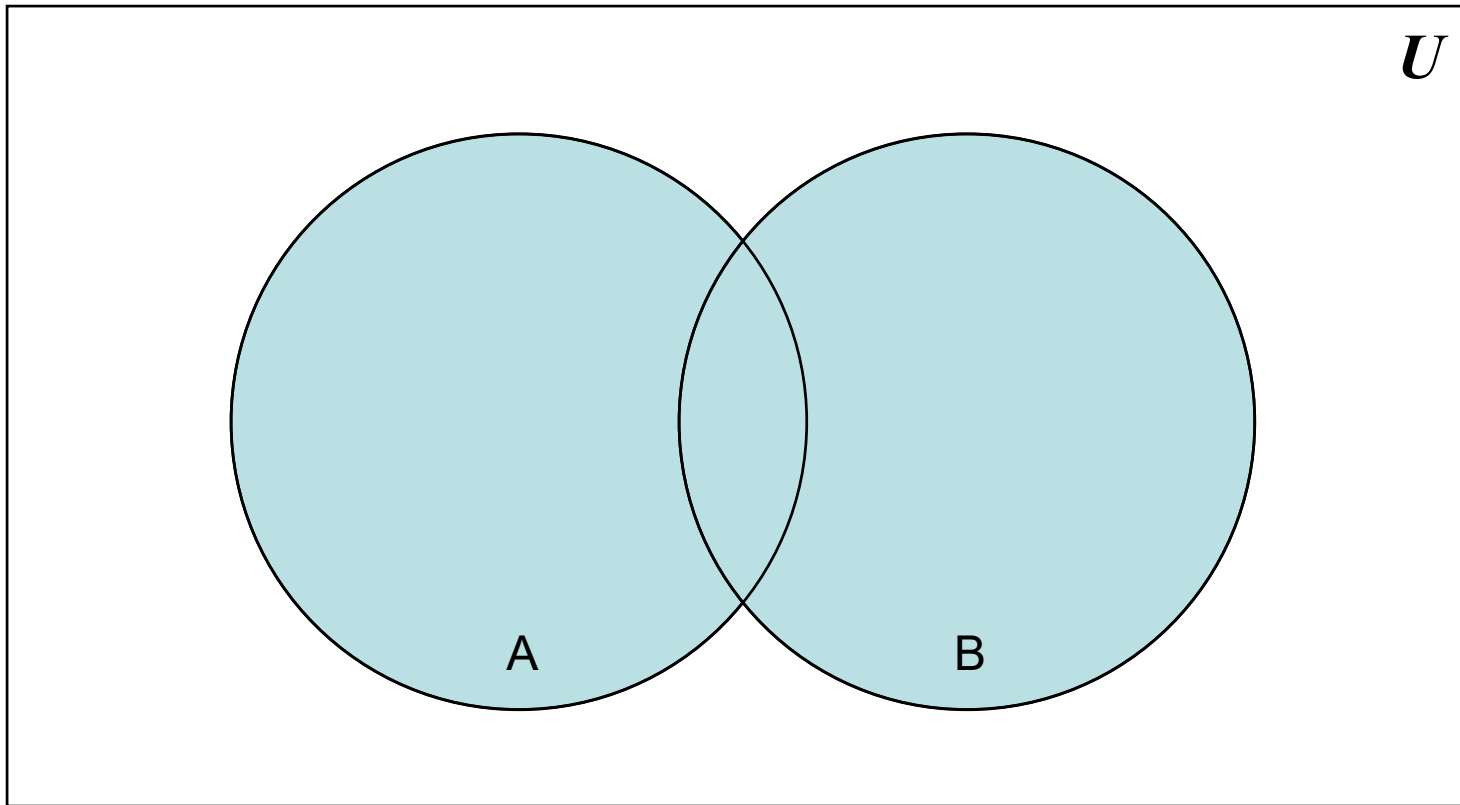


- A union of the sets contains all the elements in EITHER set

- Union symbol is usually a U
 - Example:
 $C = M \cup P$

Set operations: Union 2

$A \cup B$



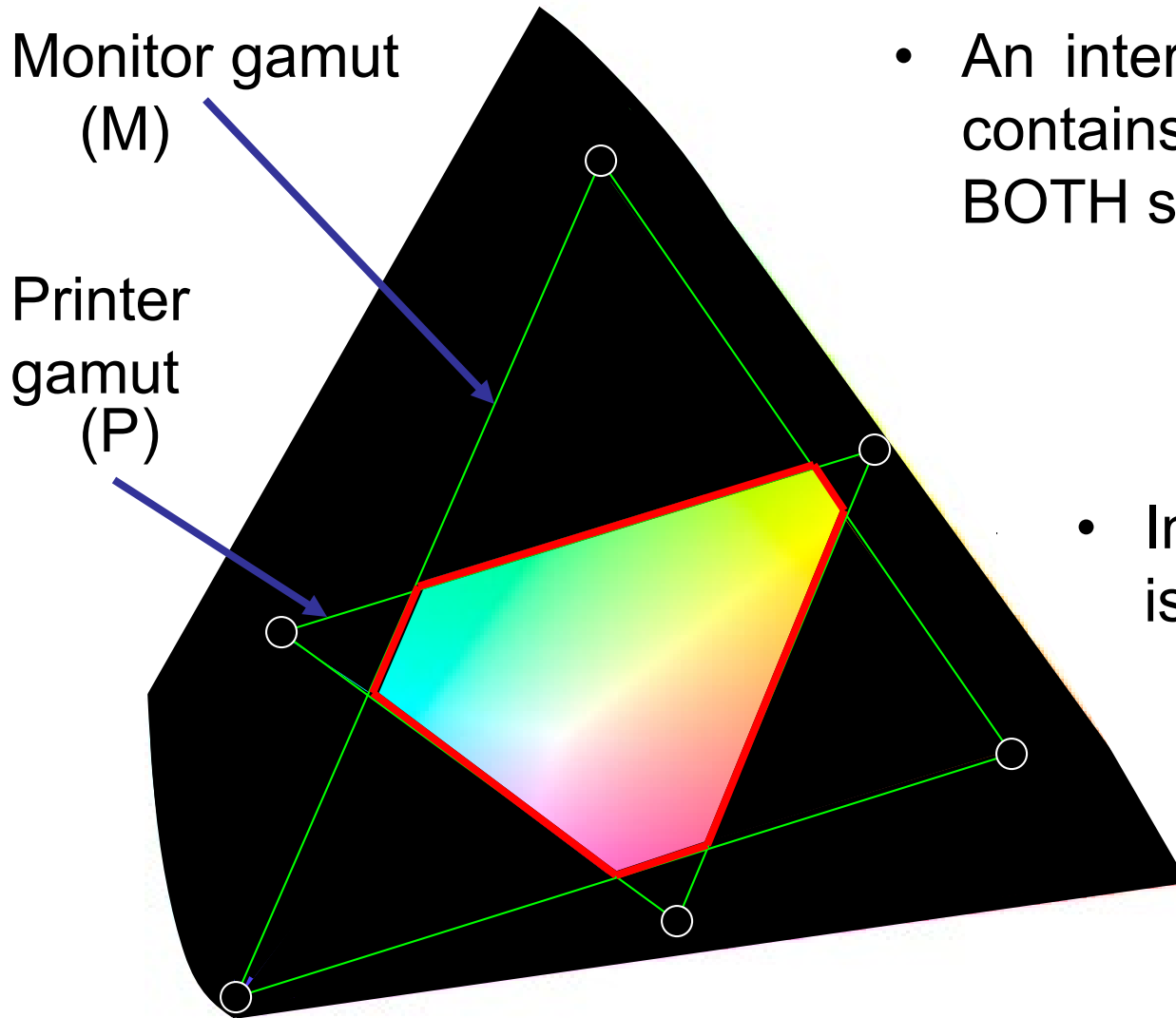
Set operations: Union 3

- Formal definition for the union of two sets:
 $A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$
- Further examples
 - $\{1, 2, 3\} \cup \{3, 4, 5\} = \{1, 2, 3, 4, 5\}$
 - $\{\text{New York, Washington}\} \cup \{3, 4\} = \{\text{New York, Washington, 3, 4}\}$
 - $\{1, 2\} \cup \emptyset = \{1, 2\}$

Set operations: Union 4

- Properties of the union operation
 - $A \cup \emptyset = A$ Identity law
 - $A \cup U = U$ Domination law
 - $A \cup A = A$ Idempotent law
 - $A \cup B = B \cup A$ Commutative law
 - $A \cup (B \cup C) = (A \cup B) \cup C$ Associative law

Set operations: Intersection 1

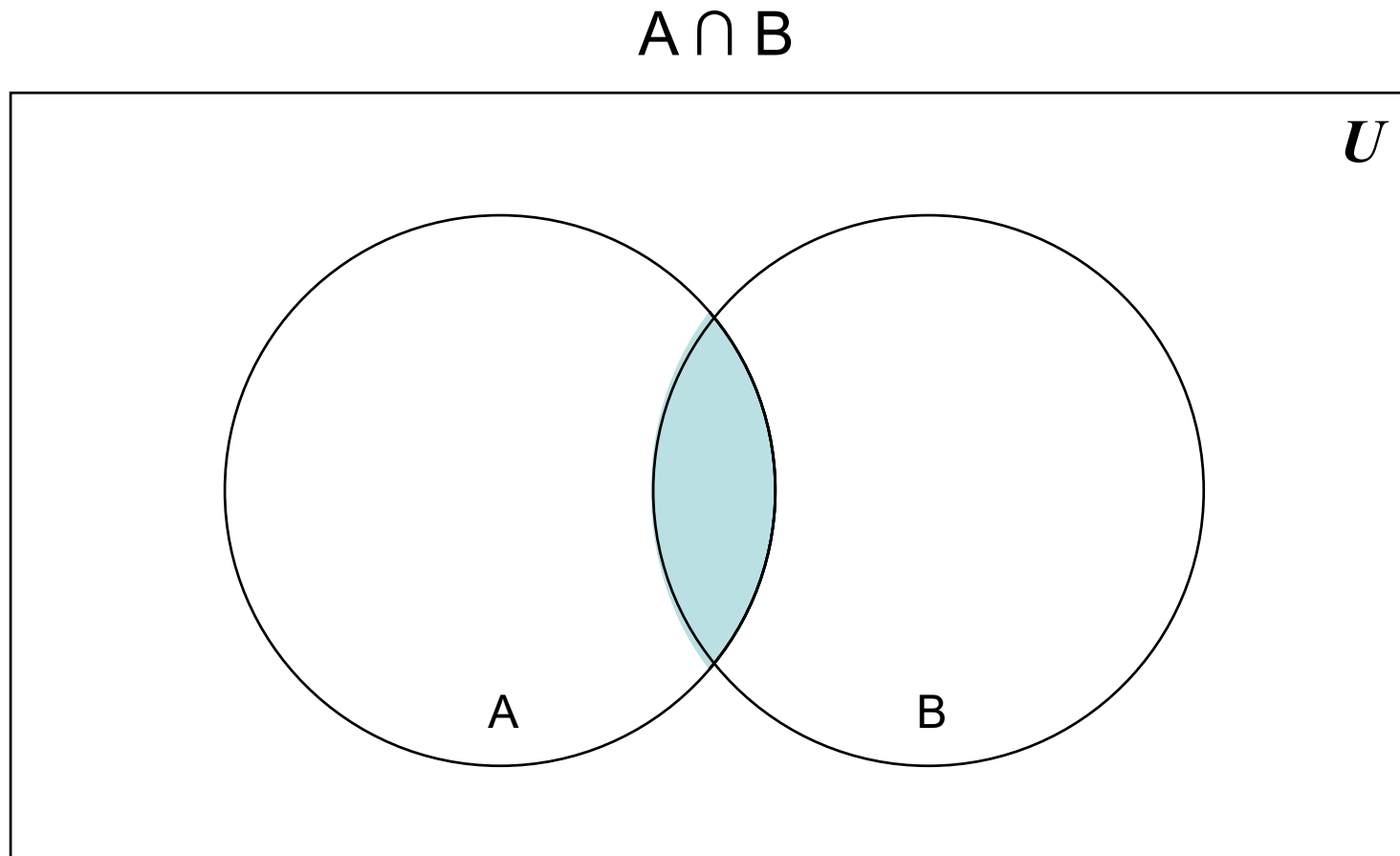


- An intersection of the sets contains all the elements in BOTH sets

- Intersection symbol is a $\cap$

- Example:
 $C = M \cap P$

Set operations: Intersection 2



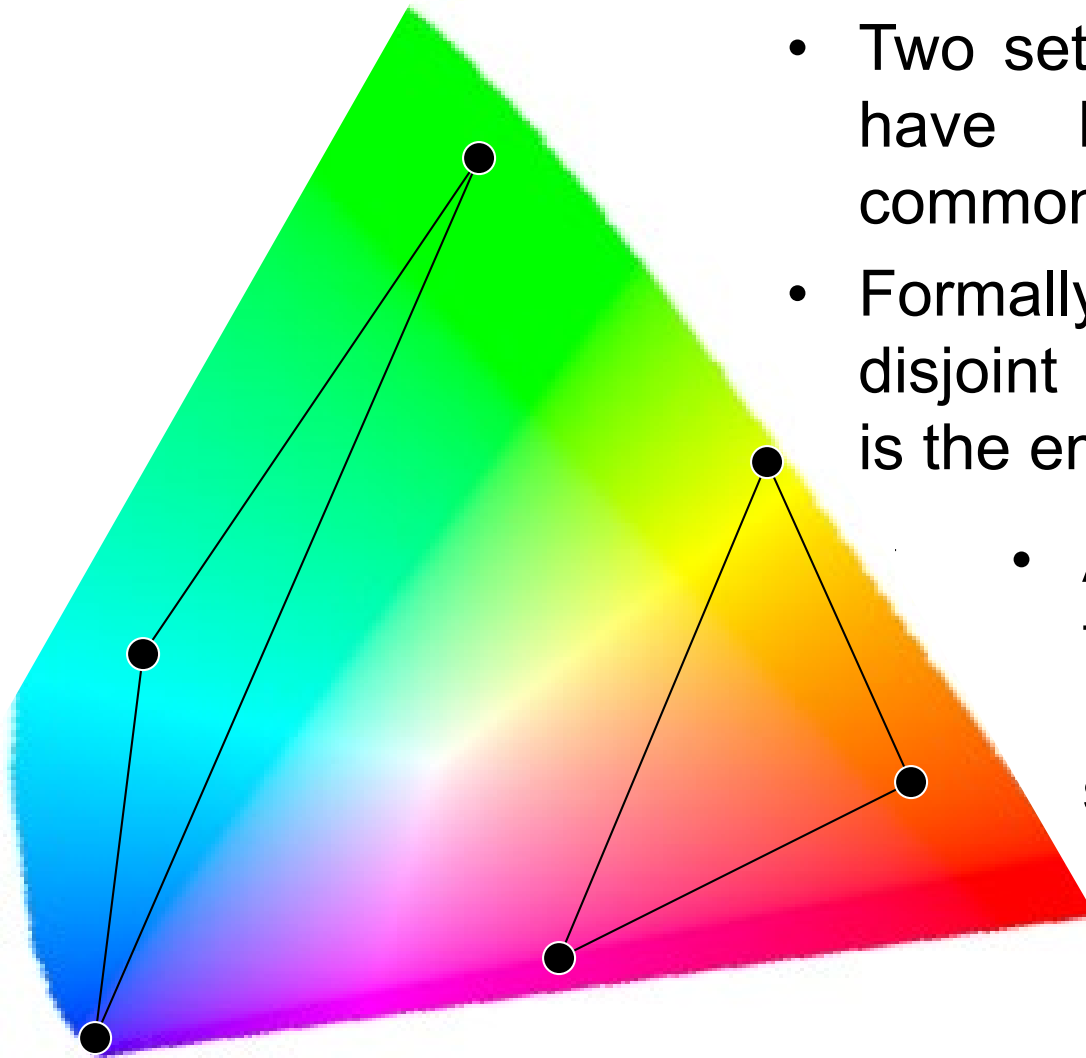
Set operations: Intersection 3

- Formal definition for the intersection of two sets: $A \cap B = \{ x \mid x \in A \text{ and } x \in B \}$
- Further examples
 - $\{1, 2, 3\} \cap \{3, 4, 5\} = \{3\}$
 - $\{\text{New York, Washington}\} \cap \{3, 4\} = \emptyset$
 - No elements in common
 - $\{1, 2\} \cap \emptyset = \emptyset$
 - Any set intersection with the empty set yields the empty set

Set operations: Intersection 4

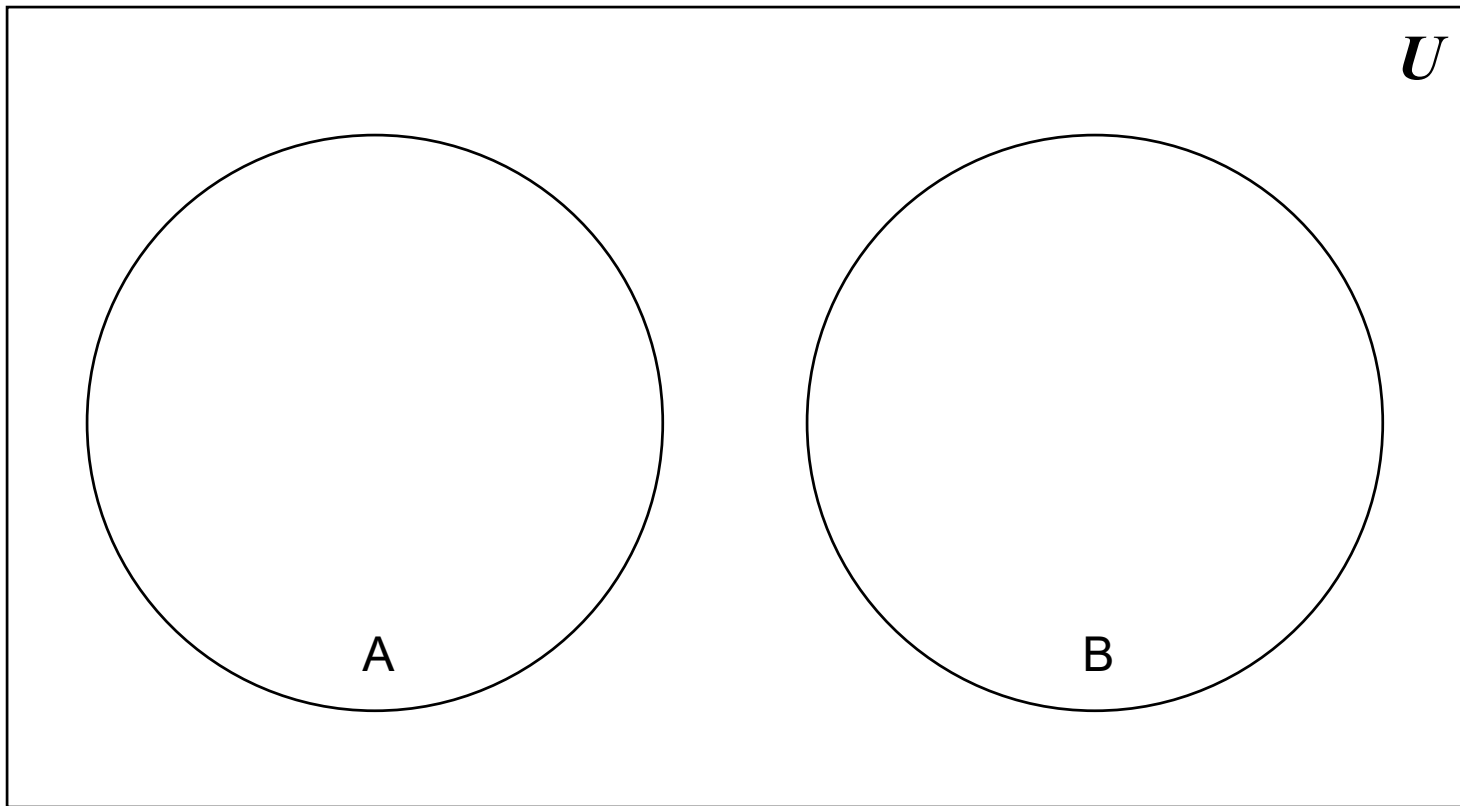
- Properties of the intersection operation
 - $A \cap U = A$ Identity law
 - $A \cap \emptyset = \emptyset$ Domination law
 - $A \cap A = A$ Idempotent law
 - $A \cap B = B \cap A$ Commutative law
 - $A \cap (B \cap C) = (A \cap B) \cap C$ Associative law

Disjoint sets 1



- Two sets are disjoint if they have NO elements in common
- Formally, two sets are disjoint if their intersection is the empty set
- Another example: the set of the even numbers and the set of the odd numbers

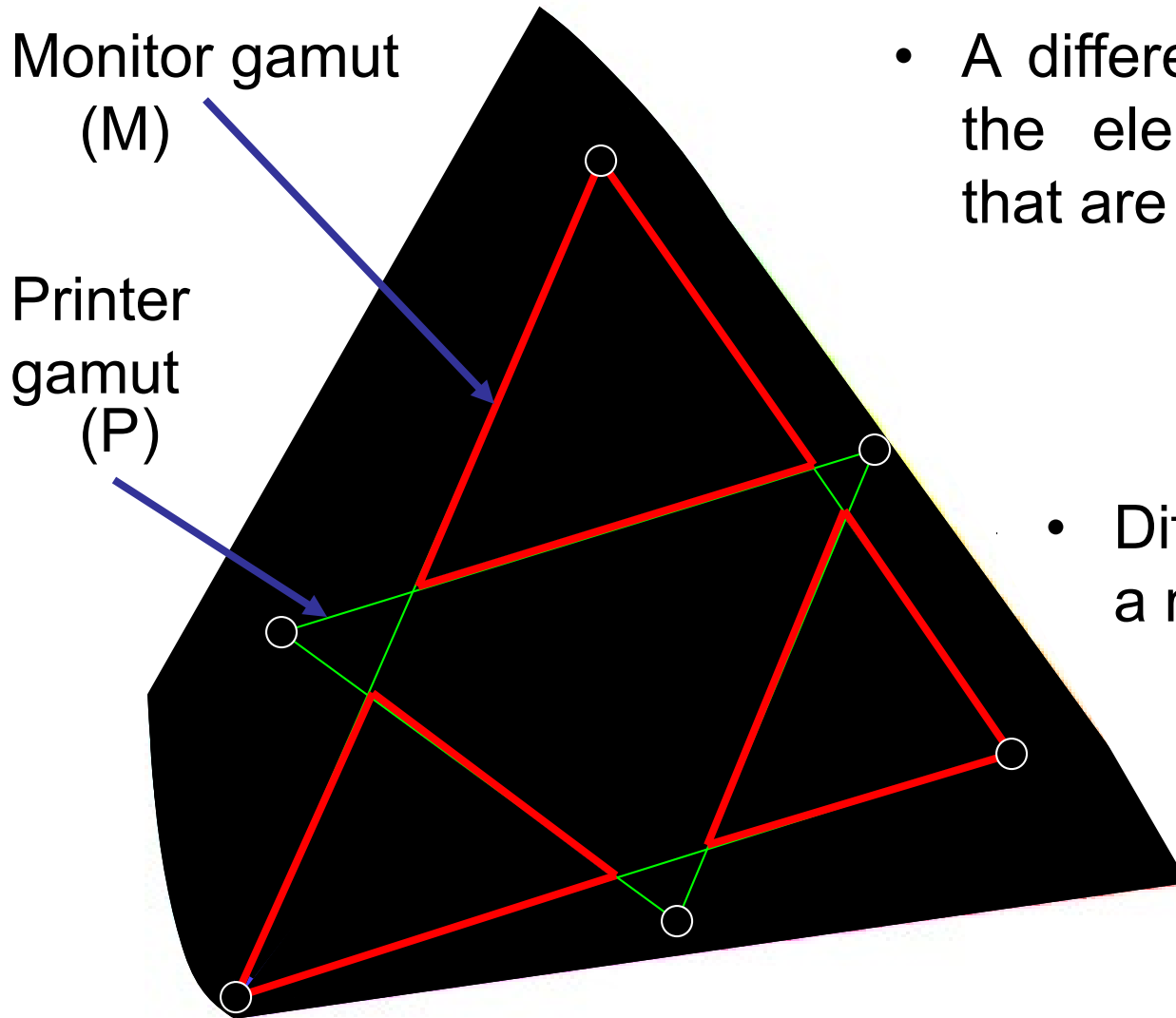
Disjoint sets 2



Disjoint sets 3

- Formal definition for disjoint sets: two sets are disjoint if their intersection is the empty set
- Further examples
 - $\{1, 2, 3\}$ and $\{3, 4, 5\}$ are not disjoint
 - $\{\text{New York, Washington}\}$ and $\{3, 4\}$ are disjoint
 - $\{1, 2\}$ and $\emptyset$ are disjoint
 - Their intersection is the empty set
 - $\emptyset$ and $\emptyset$ are disjoint!
 - Their intersection is the empty set

Set operations: Difference 1



- A difference of two sets is the elements in one set that are NOT in the other

- Difference symbol is a minus sign

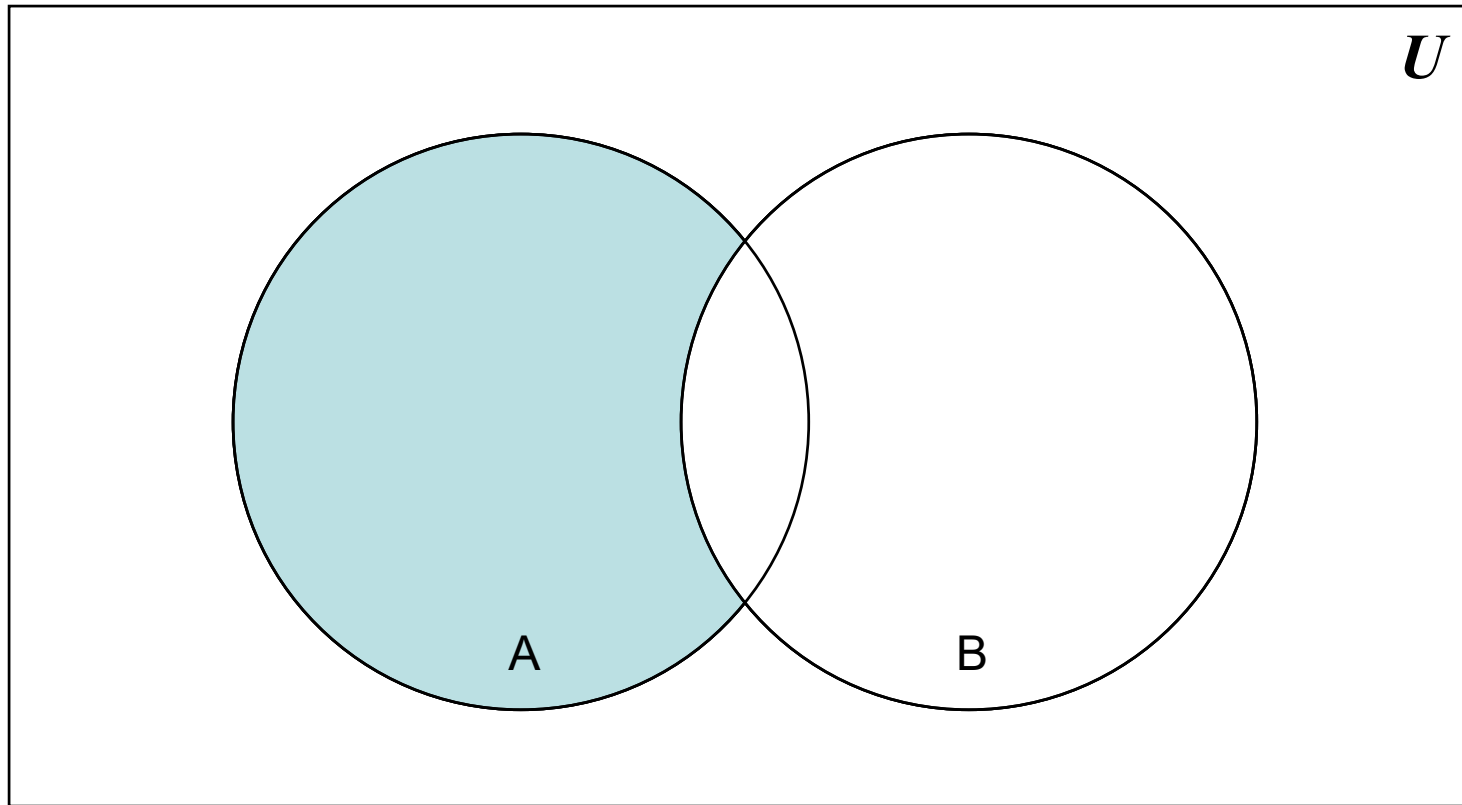
- Example:

$$C = M - P$$

- Also visa-versa:
 $C = P - M$

Set operations: Difference 2

$$B - A$$



Set operations: Difference 3

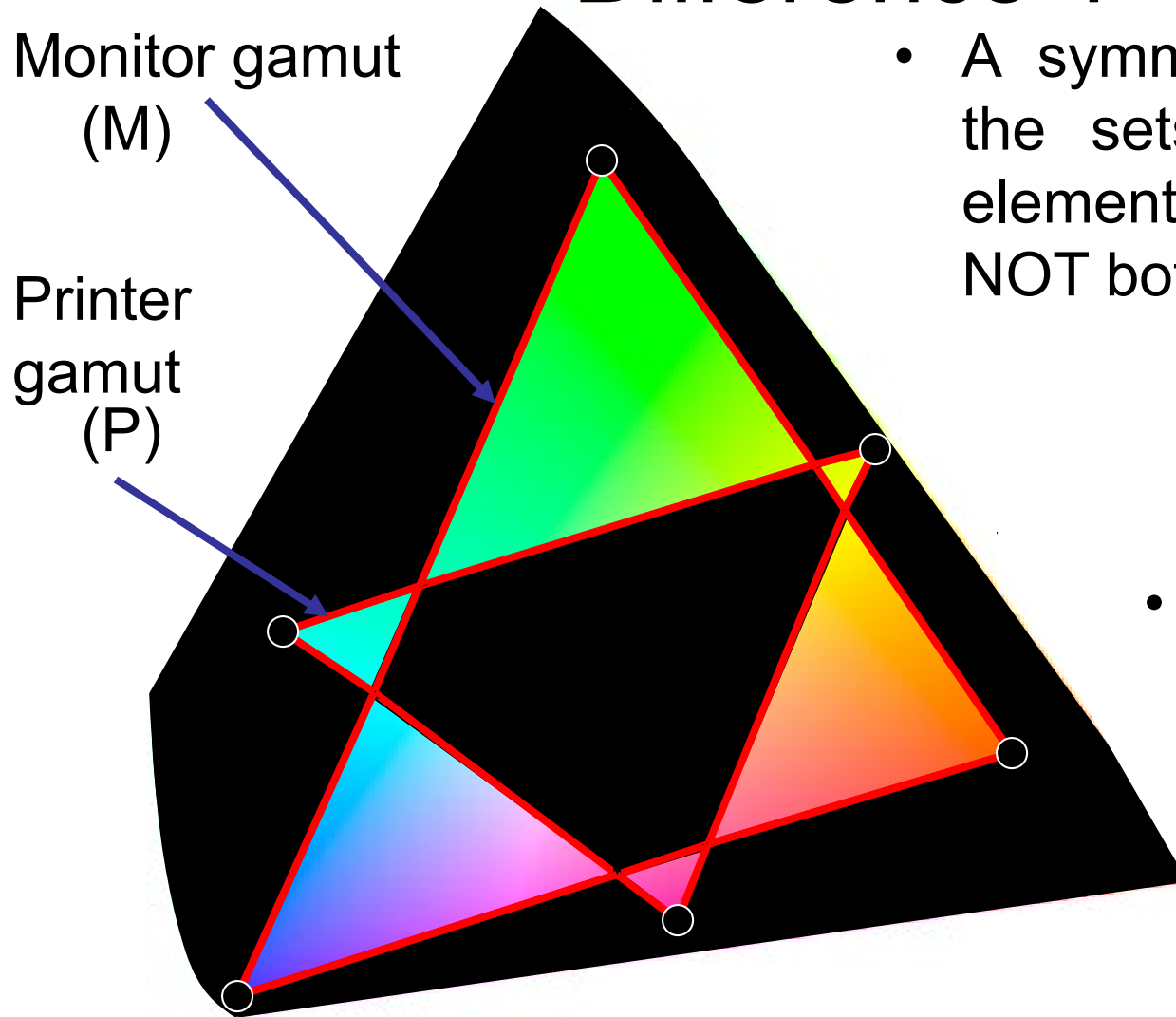
- Formal definition for the difference of two sets:

$$A - B = \{ x \mid x \in A \text{ and } x \notin B \}$$

$$A - B = A \cap \bar{B} \quad \leftarrow \text{Important!}$$

- Further examples
 - $\{1, 2, 3\} - \{3, 4, 5\} = \{1, 2\}$
 - $\{\text{New York, Washington}\} - \{3, 4\} = \{\text{New York, Washington}\}$
 - $\{1, 2\} - \emptyset = \{1, 2\}$
 - The difference of any set S with the empty set will be the set S

Set operations: Symmetric Difference 1



- A symmetric difference of the sets contains all the elements in either set but NOT both

- Symmetric diff. symbol is a $\oplus$
 - Example:
 $C = M \oplus P$

Set operations: Symmetric Difference 2

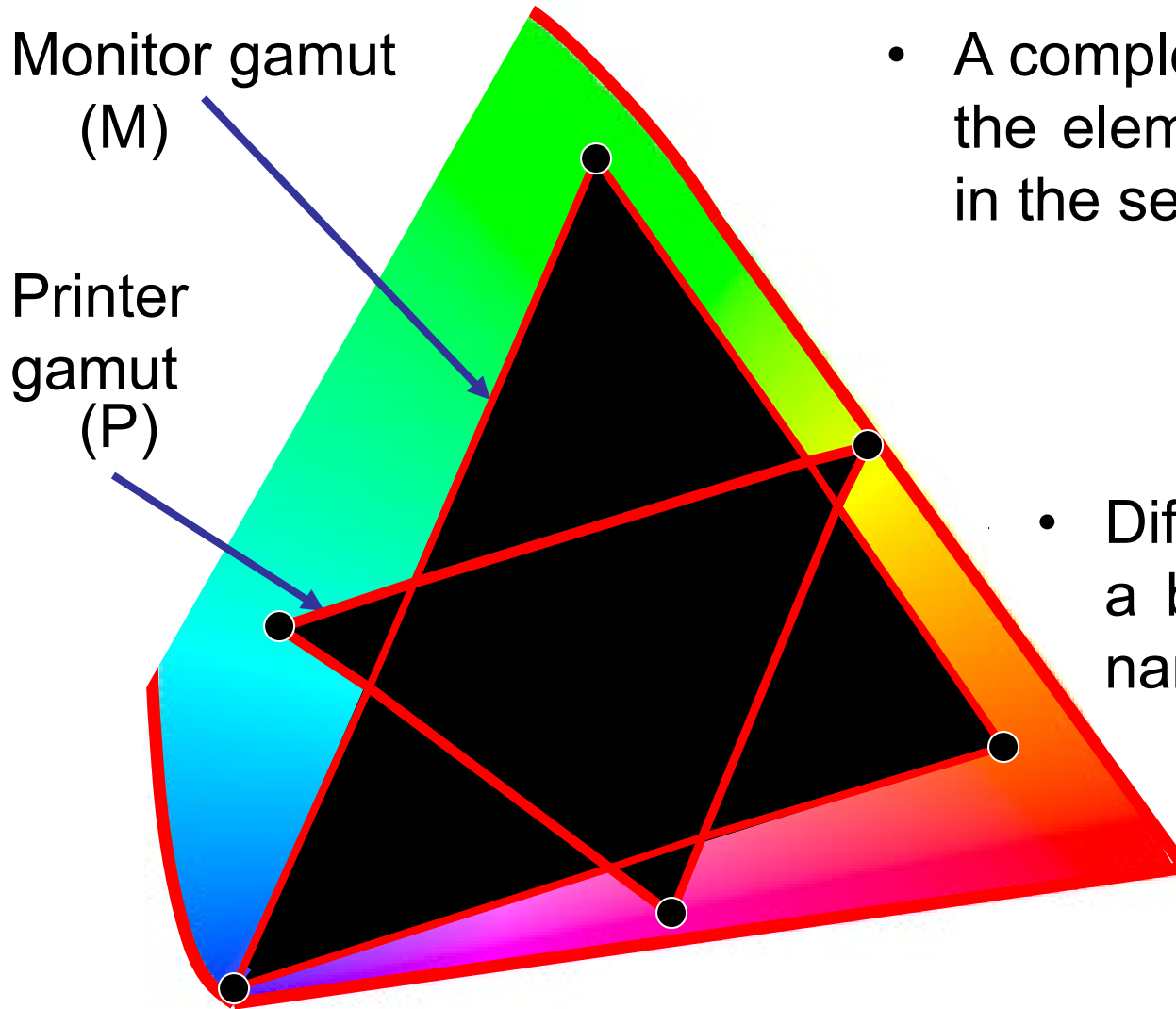
- Formal definition for the symmetric difference of two sets:

$$A \oplus B = \{ x \mid (x \in A \text{ or } x \in B) \text{ and } x \notin A \cap B \}$$

$$A \oplus B = (A \cup B) - (A \cap B) \quad \leftarrow \text{Important!}$$

- Further examples
 - $\{1, 2, 3\} \oplus \{3, 4, 5\} = \{1, 2, 4, 5\}$
 - $\{\text{New York, Washington}\} \oplus \{3, 4\} = \{\text{New York, Washington, 3, 4}\}$
 - $\{1, 2\} \oplus \emptyset = \{1, 2\}$
 - The symmetric difference of any set S with the empty set will be the set S

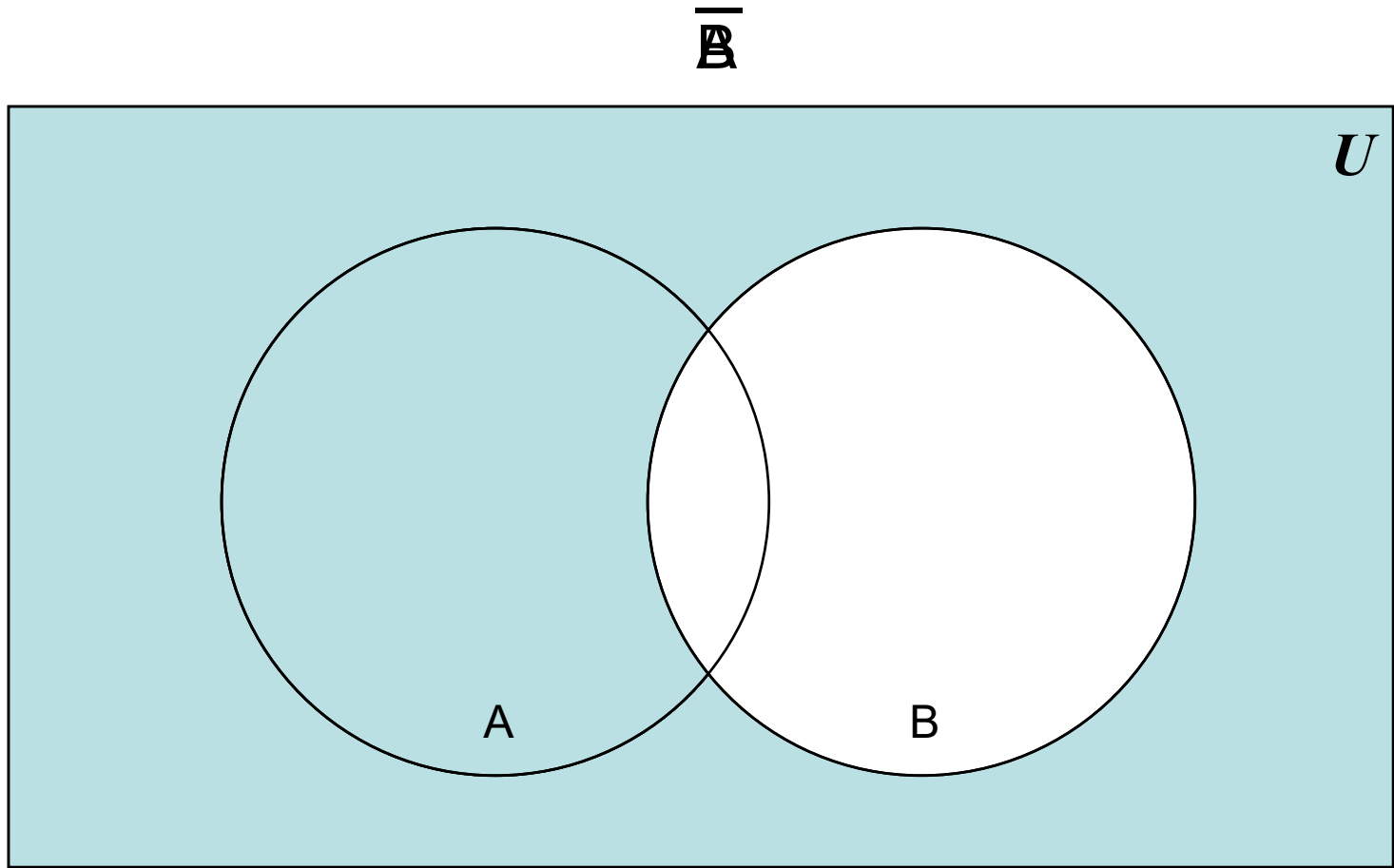
Complement sets 1



- A complement of a set is all the elements that are NOT in the set

- Difference symbol is a bar above the set name: $\bar{P}$ or $\bar{M}$

Complement sets 2



Complement sets 3

- Formal definition for the complement of a set: $\overline{A} = \{ x \mid x \notin A \}$
 - Or $U - A$, where U is the universal set
- Further examples (assuming $U = \mathbf{Z}$)
 - $\{1, 2, 3\} = \{ \dots, -2, -1, 0, 4, 5, 6, \dots \}$
 - $\{\text{New York, Washington}\} - \{3, 4\} = \{\text{New York, Washington}\}$
 - $\{1, 2\} - \emptyset = \{1, 2\}$
 - The difference of any set S with the empty set will be the set S

Complement sets 4

- Properties of complement sets

$$- \bar{\bar{A}} = A$$

Complementation law

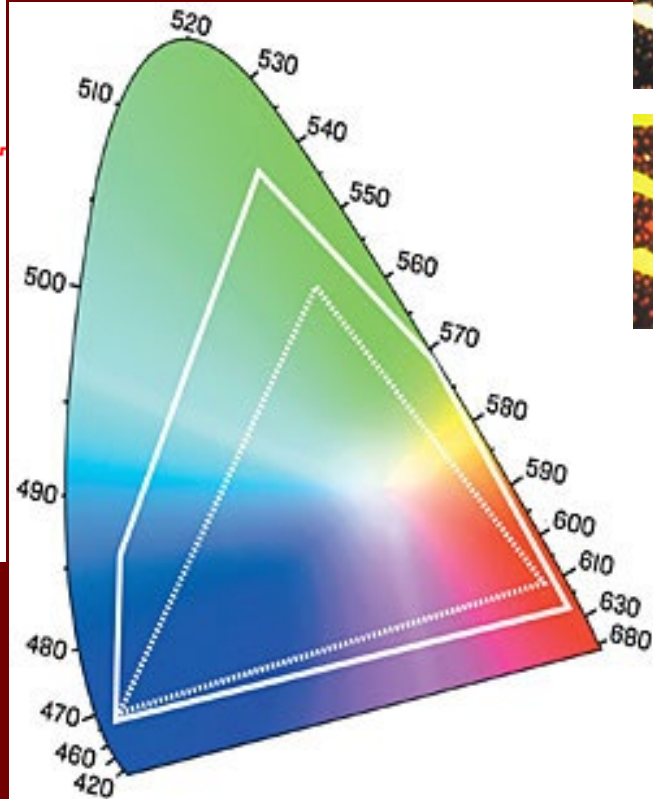
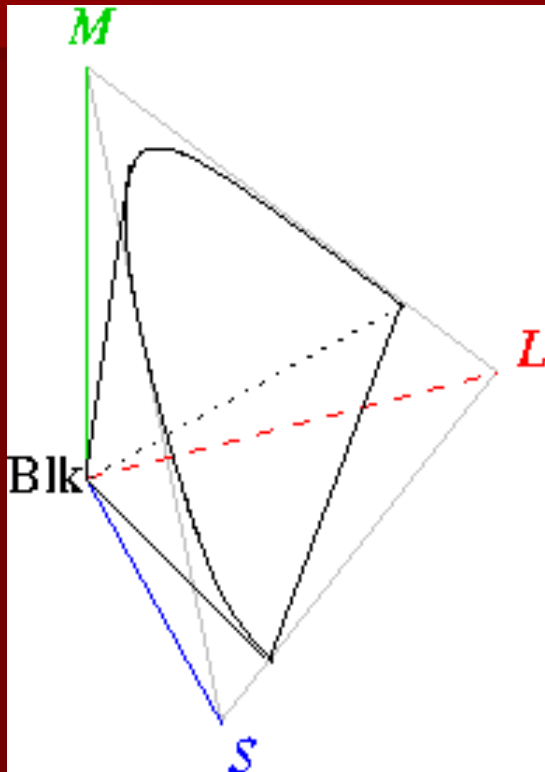
$$- A \cup \bar{A} = U$$

Complement law

$$- A \cap \bar{A} = \emptyset$$

Complement law

A last bit of color...



Set identities

- Set identities are basic laws on how set operations work
 - Many have already been introduced on previous slides
- Just like logical equivalences!
 - Replace U with $\vee$
 - Replace $\cap$ with $\wedge$
 - Replace $\emptyset$ with F
 - Replace $\mathcal{U}$ with T
- Full list on Rosen, page 89

Set identities: DeMorgan again

- These should look very familiar...

$$\overline{A \cap B} = \overline{A} \cup \overline{B}$$

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

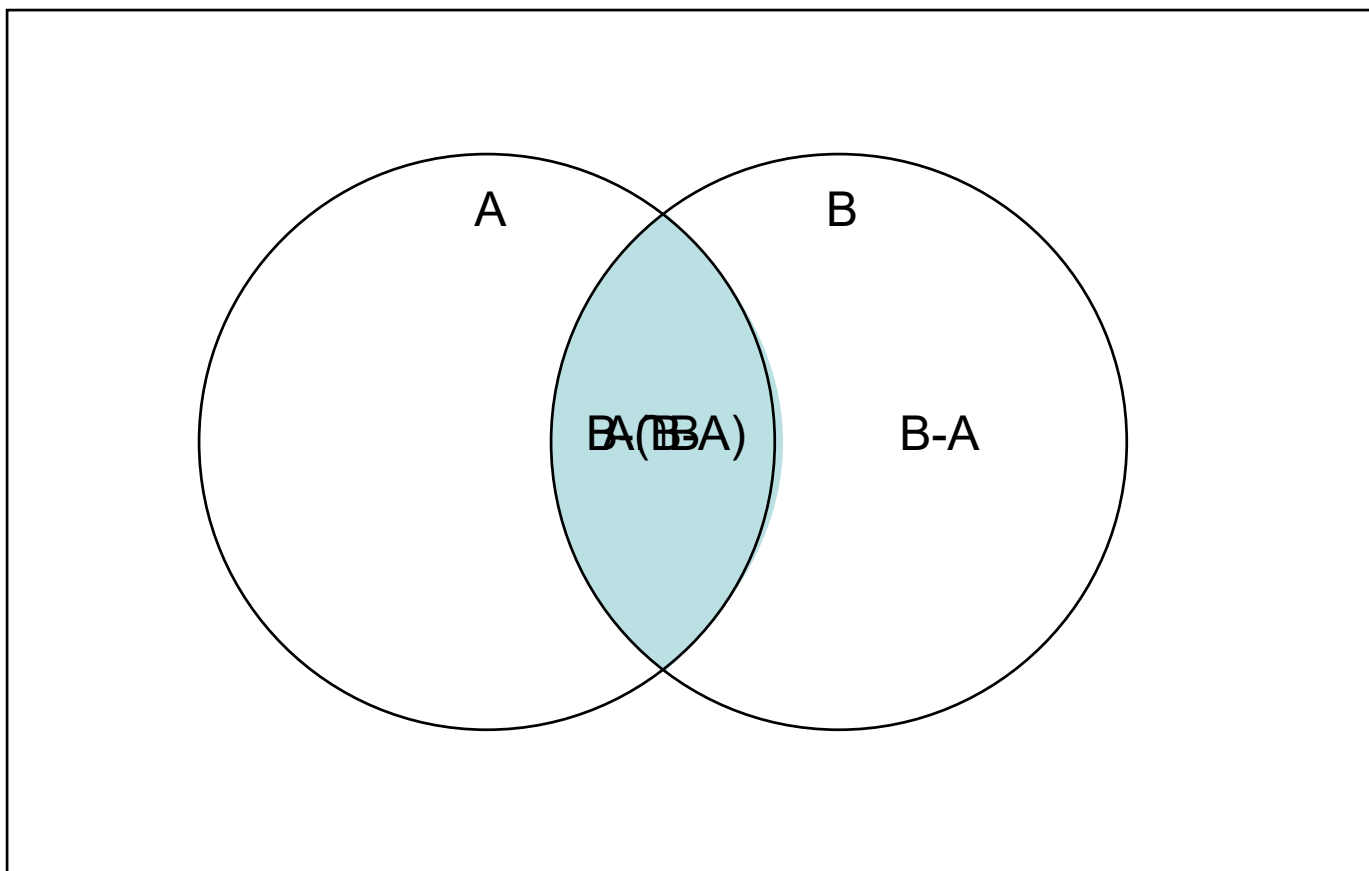


How to prove a set identity

- For example: $A \cap B = B - (B - A)$
- Four methods:
 - Use the basic set identities (Rosen, p. 89)
 - Use membership tables
 - Prove each set is a subset of each other
 - This is like proving that two numbers are equal by showing that each is less than or equal to the other
 - Use set builder notation and logical equivalences

What we are going to prove...

$$A \cap B = B - (B - A)$$



Proof by using basic set identities

- Prove that $A \cap B = B - (B - A)$

$$A \cap B = B - (B \cap \bar{A})$$

$$= B \cap \overline{(B \cap \bar{A})}$$

$$= B \cap (\bar{B} \cup \bar{\bar{A}})$$

$$= B \cap (\bar{B} \cup A)$$

$$= (B \cap \bar{B}) \cup (B \cap A)$$

$$= \emptyset \cup (B \cap A)$$

$$= (B \cap A)$$

$$= A \cap B$$

Definition of difference

Definition of difference

DeMorgan's law

Complementation law

Distributive law

Complement law

Identity law

Commutative law

What is a membership table

- Membership tables show all the combinations of sets an element can belong to
 - 1 means the element belongs, 0 means it does not
- Consider the following membership table:

A	B	$A \cup B$	$A \cap B$	$A - B$
1	1	1	1	0
1	0	1	0	1
0	1	1	0	0
0	0	0	0	0

- The top row is all elements that belong to both sets A and B
 - Thus, these elements are in the intersection of the two sets

Proof by membership tables

- The following membership table shows that $A \cap B = B - (B - A)$

A	B	$A \cap B$	$B - A$	$B - (B - A)$
1	1	1	0	1
1	0	0	0	0
0	1	0	1	0
0	0	0	0	0

- Because the two indicated columns have the same values, the two expressions are identical
- This is similar to Boolean logic!

Proof by showing each set is a subset of the other 1

- Assume that an element is a member of one of the identities
 - Then show it is a member of the other
- Repeat for the other identity
- We are trying to show:
 - $(x \in A \cap B \rightarrow x \in B - (B - A)) \wedge (x \in B - (B - A) \rightarrow x \in A \cap B)$
 - This is the biconditional!
- Not good for long proofs
- Basically, it's an English run-through of the proof

Proof by showing each set is a subset of the other 2

- Assume that $x \in B - (B - A)$
 - By definition of difference, we know that $x \in B$ and $x \notin B - A$
- Consider $x \notin B - A$
 - If $x \in B - A$, then (by definition of difference) $x \in B$ and $x \notin A$
 - Since $x \notin B - A$, then only one of the inverses has to be true (DeMorgan's law): $x \notin B$ or $x \in A$
- So we have that $x \in B$ and $(x \notin B \text{ or } x \in A)$
 - It cannot be the case where $x \in B$ and $x \notin B$
 - Thus, $x \in B$ and $x \in A$
 - This is the definition of intersection
- Thus, if $x \in B - (B - A)$ then $x \in A \cap B$

Proof by showing each set is a subset of the other 3

- Assume that $x \in A \cap B$
 - By definition of intersection, $x \in A$ and $x \in B$
- Thus, we know that $x \notin B - A$
 - $B - A$ includes all the elements in B that are also not in A not include any of the elements of A (by definition of difference)
- Consider $B - (B - A)$
 - We know that $x \notin B - A$
 - We also know that if $x \in A \cap B$ then $x \in B$ (by definition of intersection)
 - Thus, if $x \in B$ and $x \notin B - A$, we can restate that (using the definition of difference) as $x \in B - (B - A)$
- Thus, if $x \in A \cap B$ then $x \in B - (B - A)$

Proof by set builder notation and logical equivalences 1

- First, translate both sides of the set identity into set builder notation
- Then massage one side (or both) to make it identical to the other
 - Do this using logical equivalences

Proof by set builder notation and logical equivalences 2

$B - (B - A)$	Original statement
$= \{x \mid x \in B \wedge x \notin (B - A)\}$	Definition of difference
$= \{x \mid x \in B \wedge \neg(x \in (B - A))\}$	Negating “element of”
$= \{x \mid x \in B \wedge \neg(x \in B \wedge x \notin A)\}$	Definition of difference
$= \{x \mid x \in B \wedge (x \notin B \vee x \in A)\}$	DeMorgan’s Law
$= \{x \mid (x \in B \wedge x \notin B) \vee (x \in B \wedge x \in A)\}$	Distributive Law
$= \{x \mid (x \in B \wedge \neg(x \in B)) \vee (x \in B \wedge x \in A)\}$	Negating “element of”
$= \{x \mid F \vee (x \in B \wedge x \in A)\}$	Negation Law
$= \{x \mid x \in B \wedge x \in A\}$	Identity Law
$= A \cap B$	Definition of intersection

Proof by set builder notation and logical equivalences 3

- Why can't you prove it the “other” way?
 - I.e. massage $A \cap B$ to make it look like $B - (B - A)$
- You can, but it's a bit annoying
 - In this case, it's not simplifying the statement

Computer representation of sets 1

- Assume that U is finite (and reasonable!)
 - Let U be the alphabet
- Each bit represents whether the element in U is in the set
- The vowels in the alphabet:
abcdefghijklmnopqrstuvwxyz
10001000100000100000100000
- The consonants in the alphabet:
abcdefghijklmnopqrstuvwxyz
01110111011111011111011111

Computer representation of sets 2

- Consider the union of these two sets:

$$\begin{array}{r} 10001000100000100000100000 \\ \vee \underline{01110111011111011111011111} \\ 11111111111111111111111111 \end{array}$$

- Consider the intersection of these two sets:

$$\begin{array}{r} 10001000100000100000100000 \\ \wedge \underline{01110111011111011111011111} \\ 00000000000000000000000000 \end{array}$$

Rosen, section 1.7 question 14

- Let A , B , and C be sets. Show that:
 - a) $(A \cup B) \subseteq (A \cup B \cup C)$
 - b) $(A \cap B \cap C) \subseteq (A \cap B)$
 - c) $(A - B) - C \subseteq A - C$
 - d) $(A - C) \cap (C - B) = \emptyset$